



PennState

College of Information
Sciences and Technology

Lab Assignment Report

COURSE:	IST 894
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General Context

The course, “Cybersecurity Senior Management and Information Security Governance (ISG)” speaks to the reality that cybersecurity is not only a technical concern, but a key leadership responsibility that requires involvement from senior leadership. Gone are the days of hiring an IT staff and leaving them to their own devices to “do IT.” From healthcare to financial institutions, cybersecurity is recognized as a strategic imperative (Nocera et al., 2025). In this course, governance was defined as the system of practices, controls, and procedures that align with the organization’s mission, vision, and legal obligations while meeting the needs of the stakeholders. That governance structure is important in the execution of cybersecurity requirements in an organization.

Cybersecurity leadership goes far beyond a Chief Information Security Officer (CISO). While a CISO is responsible for setting the cybersecurity strategy and the development of cybersecurity policies (U.S. Chief Information Officer’s Council, n.d.), others must be involved. The course stressed the importance of cross-functional committees, e.g., finance, compliance, operations, and human resources, to ensure cybersecurity is integrated into every aspect of the business. Real risk management is not just an isolated IT issue. It is a strategic enterprise issue and should be treated as such.

Another important takeaway from this course was the importance of administrative, physical, and technical controls. The administrative controls cover things like policies and training. The physical controls cover aspects such as facility security and access control measures. The technical controls are the items that are generally seen as IT responsibilities; things like firewalls, encryption of data, and multi-factor authentication, to name a few. This really drives home the point that cybersecurity is an organizational responsibility. Frameworks like ISO 27001, NIST frameworks, and COBIT are great places to start and are foundational for strong cybersecurity.

Technical Context

This course provided a structured approach for the development and maintenance of a strong cybersecurity program from the strategic leadership perspective. Something the instructor spoke about often, whether directly or indirectly, was the importance of continuously identifying risks, aligning controls, and implementing them effectively, all through successful governance. Established best practices and frameworks can be used as a guide for successful implementation.

The role of the CISO was explored in great detail, even going as far as separating CISOs out into three specific categories: the Technical-oriented CISO, Strategically-oriented CISO, and the Business Policy-oriented CISO. Mozzola (2023) speaks to the rapidly evolving nature of the CISO and states that the purely technical CISO is obsolete. Business strategy is a much larger piece of the pie now. The course echoed this by consistently referring to leadership styles and the need for collaboration across the entire enterprise, not just IT-focused workspaces.

The CIA (confidentiality, integrity, and availability) triad is a well-known pillar of cybersecurity that is used as a model to guide policy development within an organization (Hashemi-Pour & Chai, 2023). As such, it was framed as a practical policy design guide within this course. The security, trust, and stability (STS) model was described as the binding that brings the cybersecurity management side of the CIA together with the strategic leadership side of the organization. Finally, a key takeaway for successful cybersecurity leadership is that strong leadership will develop cybersecurity controls that build stability. On the administrative side, that may look like a well-developed and executable business continuity plan. The physical side may see reliable air handling with generators backing them up. The technical side may involve redundant cloud nodes hosting business-critical data. Whatever the case may be, stability through cybersecurity is the goal.

Screenshots



Figure 1: Certificate of Completion

References

- Hashemi-Pour, C., & Chai, W. (2023, December 21). *What is the CIA triad (confidentiality, integrity and availability)?* WhatIs. <https://www.techtarget.com/whatis/definition/Confidentiality-integrity-and-availability-CIA>
- Mazzola, C. (2023, December 19). *The future of CISO: From technical expert to business leaders*. PECB Insights. <https://insights.pecb.com/future-ciso-from-technicalexpert-business-leaders/>
- Nocera, J., Gorham, M., & Horton, M. (2025, March 17). *Cybersecurity Investments: a strategic imperative for modern business leadership*. Financial Executives International. Retrieved August 9, 2025, from <https://www.financialexecutives.org/FEI-Daily/March-2025/Cybersecurity-Investments-A-Strategic-Imperative-f.aspx>
- U.S. Chief Information Officer's Council. (n.d.). *Chief Information Security Officer (CISO)*. CIO.GOV. Retrieved August 9, 2025, from <https://www.cio.gov/handbook/key-stakeholders/ciso/>